REMOTE MONITORING
Of Cardiac Devices

Rapid progress in information technology and medical sciences has resulted in several implantable devices. This article focuses on the heart and related devices, which are used to provide cardiac resynchronisation therapy, help maintain rhythm, and prevent sudden cardiac death

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The focus in personal healthcare has shifted from hospital care to home care and ultimately to healthcare on-the-move. As a result, the usage and popularity is rapidly shifting from fixed medical devices and systems to portable ones and ultimately to the wearable devices. The first-generation devices needed user involvement to obtain and record the data that was processed offline—for example, the heart holter used for monitoring and recording cardiac activity.

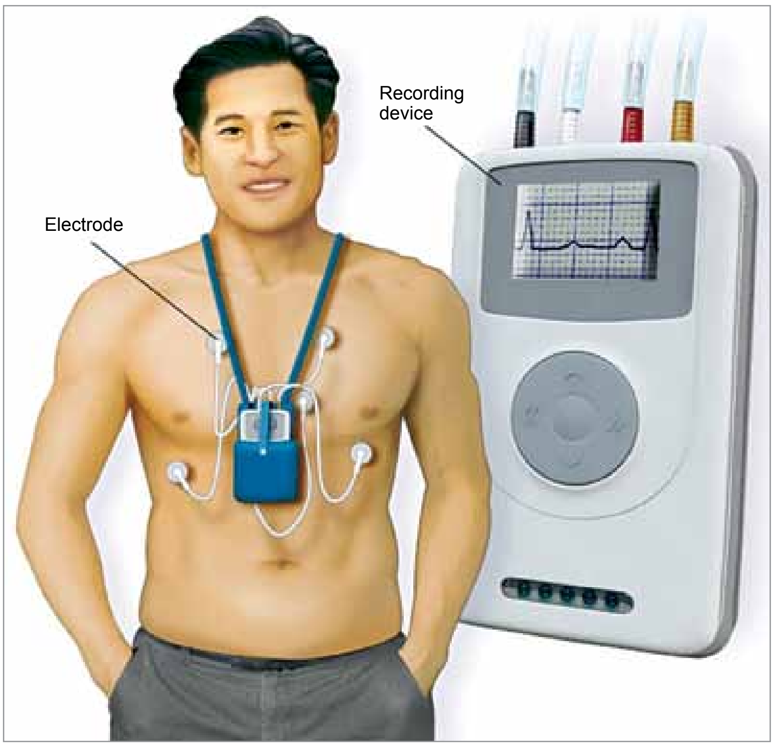

Fig. 1: Heart holter, a wearable device that keeps track of heart rhythm (Source: www.mayoclinic.org)

Rapid progress in information technology and medical sciences has resulted in several implantable devices. This article focuses on the heart and related devices. Implanted electronic cardiac devices (IECDs) include a broad spectrum of devices, such as pacemakers (PMs), implantable cardioverter defibrillators (ICDs), cardiac resynchronisation therapy (CRT) devices, implantable loop recorders (ILRs), and implantable hemodynamic monitors (IHMs). These devices are used to provide cardiac resynchronisation therapy, help maintain rhythm, and prevent sudden cardiac death (see box).

The heart of all these devices is a microprocessor with a miniature onboard power source. The microprocessor is programmed to collect data 24 × 7 and initiate the necessary action when it detects an event. For example, if the heart starts beating too fast, the device delivers a therapy to restore the normal heart rhythm.

The device, when implanted in a patient, needs regular monitoring. The patient has to make visits to the healthcare centre every three to six months to enable the healthcare team to check the device. During the visits, the data recorded by the device is downloaded to provide information regarding the significant events that have occurred, such as device malfunction, arrhythmia, changes in hemodynamic status or inadvertent changes in programmed parameters. This enables assessing the effects of the therapy as a part of a complete cardiac evaluation and optimisation of the device function.